

Code No: 54013**JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD****B.Tech II Year II Semester Examinations, July/August - 2021****DATABASE MANAGEMENT SYSTEMS****(Common to CSE, IT)****Time: 3 hours****Max. Marks: 75**

Answer any five questions
All questions carry equal marks

- 1.a) Describe the database system structure with a neat diagram.
- b) How does a query get processed in query processor? Demonstrate with an example. [7+8]
2. What is the significance of class hierarchies in ER data model? Give illustrations. [15]
- 3.a) What is a view? Discuss its role in database security.
- b) Differentiate between tuple relational calculus with domain relational calculus. [7+8]
- 4.a) Compare nested query and correlated query. Discuss suitable examples.
- b) What is a trigger? Why is it needed in DBMS? Explain various types of triggers. [7+8]
5. Describe multi valued dependencies and the normal form associated with it. [15]
6. Discuss about the performance of locking and intention locks. [15]
- 7.a) Explain buffer management along with page replacement policies for buffer.
- b) Briefly discuss write ahead log protocol for recovery. [8+7]
8. What is an index? Differentiate between sparse and dense indices. [15]

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